

Zhanchen Dai

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Education

Télécom Paris, Diplôme d'Ingénieur (M.Eng.) Expected Sep 2026 – Jul 2028

Tracks: Signal Processing for Artificial Intelligence; Distributed Software Systems

CUHK-Shenzhen, School of Data Science, B.Eng. in CSE Sep 2022 – Jul 2026

- Major GPA: 3.909/4.0 (Rank: 7/115); admitted to the SDS's Bachelor-PhD Track
- Nanyang Technological University, GEM Trailblazer Exchange Program, Aug 2024 – Dec 2024

Research Experience

Research Intern, Xiaomi XiaoAI-Plus, Auditory Understanding Department May 2026 – Aug 2026

- **DashengLM-Spatial**: Explored architectural simplifications and compression of the model; conducted the benchmark; organized internal code into the open-sourced GitHub repo and Hugging Face release.
- **Binaural spatial MAE encoder (early-stage)**: Trained a binaural MAE encoder whose downstream classification and frame-level localization heads both clearly outperform random baselines; next step is extending the front-end to accept more channels toward an array-invariant encoder.

Research Assistant, CUHK-Shenzhen, Advisor: Prof. Benyou Wang Dec 2024 – May 2026

- **EchoX**, an end-to-end speech-to-speech LLM: designed the streaming segmentation strategy that operates directly on raw LLM embeddings, avoiding cascaded text decoding to preserve end-to-end continuity and prosody. Trained on ~6,000 hours of speech SFT data, matching open-source peers of comparable scale on several QA benchmarks.
- **MTalk-Bench**, a benchmark for multi-turn speech dialogue evaluation: led its full technical implementation, revealing systematic SLLM weaknesses and quantifying bias in LLM judges; led its open-source release.
- **EchoX-Stream**, text-free end-to-end speech generation: expanding the discrete speech token space from ~6k to ~30k via a “unit language,” with phoneme-level bridging and optimal transport alignment for direct speech-token generation. Responsible for the inference framework, online demo, and first-stage post-training.

Research Assistant, CUHK-Shenzhen, Advisor: Prof. Yixiang Fang Sep 2023 – May 2024

- Studied exact solving of maximum/maximal k -Defective Biclique problems in graph data, designing heuristic pruning strategies based on the BK algorithm, analyzing theoretical complexity, and experimentally validating pruning effectiveness.

Internship Experience

Backend Development Intern, Wenzhou Institute, Zhejiang University Jul 2023 – Aug 2023

- Designed and implemented RESTful APIs using the Spring Boot and MyBatis frameworks, supporting complex business logic and efficient database queries.

Publications

EchoX: Towards Mitigating Acoustic-Semantic Gap via Echo Training for Speech-to-Speech LLMs (submitted to ICLR 2026) · arXiv:2509.09174 ·  code ·  model

Yuhao Zhang, Yuhao Du, Zhanchen Dai, Xiangnan Ma, Kaiqi Kou, Benyou Wang, Haizhou Li

MTalk-Bench: Evaluating Speech-to-Speech Models in Multi-Turn Dialogues via Arena-style and Rubrics Protocols (submitted to IJCAI 2026) · arXiv:2508.18240 ·  code ·  dataset

Yuhao Du, Qianwei Huang, Guo Zhu, Zhanchen Dai, Shunian Chen, Qiming Zhu, Yuhao Zhang, Li Zhou, Benyou Wang

Competitions & Awards

- **Top 3**, CUHK-Shenzhen Intercollegiate Programming Contest, 2023
- CSP 2020 First Prize, NOIp 2020 Second Prize, CSP 2019 Second Prize

Skills

- **Programming Languages**: Python, C++, C, Java, SQL, Shell
- **ML Frameworks**: PyTorch, HuggingFace Transformers, PEFT (LoRA), DeepSpeed
- **Speech Processing**: G2P phoneme analysis, BPE encoding, audio feature extraction (codec), MAE
- **Languages**: Chinese (native), English (fluent), French (learning)